HANYU LIU

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RESEARCH INTERESTS

- Multi-contact motion planning
- Robot perception & control
- Reinforcement learning

EDUCATION

National University of Singapore (NUS), Singapore *Aug 2024 - Jun 2028*
Bachelor of Engineering in Electrical Engineering, Minor in Computing

- GPA: 4.92/5.00 (DEAN'S LIST)
- Core Courses: Programming Methodology(A), Electrical Engineering Principles & Practice I&II(A+), Design & Make(A), Quantitative Reasoning with Data(A-), Introduction to Machine Learning(A), Digital Logic Circuit(A), Microcontroller Programming & Interfacing, Signals & Systems, Analytical Methods in Electrical & Computing Engineering(A)

ACADEMIC PROJECT

Embedded Systems Game Platform (STM32, Interrupts & Real-Time I/O) Sep 2025 - Nov 2025

- Implemented a **non-blocking embedded system** on STM32 Cortex-M4 using EXTI and UART interrupts, supporting responsive gameplay, reliable button debounce, and mode switching.
- Optimized sensor and I/O performance via **cached calibration, EMA filtering, ring-buffered UART CLI**, and buffered OLED updates to eliminate display- and I²C-induced lag.
- Debugged and resolved **low-level timing and UART clock issues**, ensuring stable baud rates, deterministic serial output, and robust real-time behavior.
- **Tools/skills**: STM32 (HAL), ARM Cortex-M4, C, Interrupts (EXTI/UART), Ring Buffers, I²C, UART, Embedded Debugging

ARM Cortex-M4 IIR Filter (C + Assembly) *Aug 2025 - Oct 2025*

- Re-designed IIR filter from Direct Form I to Direct Form II with ring buffer, cutting memory operations by ~70% (O(N) to O(1) per sample).
- Hand-tuned assembly inner loop, reducing stack usage by ~40% and avoiding unnecessary callee-saved registers.
- Achieved 2.5× throughput improvement on STM32 Cortex-M4, lowering per-sample cycles and bus traffic while maintaining fixed-point DSP accuracy.
- **Tools/skills**: ARM Cortex-M4, STM32, Fixed-point DSP

PlanMate — Collaborative Task & Calendar Web App *May 2025 - Aug 2025*

- Developed a collaborative web app with login & authentication, task management, and progress tracking features.
- Implemented real-time group collaboration, calendar integration, and permission workflows for reliability and scalability.
- **Tools/skills**: React, TypeScript, Supabase/Postgres

Autonomous Robotic Vacuum Cleaner *Jan 2025 - Apr 2025*

- Designed sensing, navigation, and docking subsystems (IR docking beacon; randomized turning with corner-escape heuristic).
- Performed power budgeting and battery sizing; integrated motor control and safety features.
- Built dust-bag weight alert and status display; validated coverage via cleaning-effect simulation.
- **Tools/skills**: ESP32, Arduino C/C++, IR/ultrasonic/LiDAR sensing, DC motor control.

Off-Grid Solar Street Lamp *Aug 2024 - Nov 2024*

- Engineered a solar streetlamp with PWM auto-dimming using LDR sensing and debounce logic to meet target ground illumination.
- Sized PV and battery for multi-day autonomy; integrated MPPT charging and DC-DC regulation.
- Produced costed BOM and prototype verification.
- **Tools/skills**: Arduino, LDR sensing, MPPT, power electronics, LED arrays.

EG1311 Catapult Robot (Single-Motor 4-Wheel Drive) *Aug 2024 - Oct 2024*

- Built single-motor drivetrain via multi-stage gear train, optimized traction to clear obstacles and ramp.
- Implemented ultrasonic ranging and a servo-driven catapult; resolved false triggers and cable-management issues through iterative testing.
- **Tools/skills**: Arduino, Servo Control, mechanical design, laser-cut chassis & gears.

INTERNSHIP EXPERIENCE

Neusoft Ruichi Automotive Technology, Shenyang *Feb 2023 – Mar 2023*

R&D Intern

- Conducted experiments on miniature car models to study energy efficiency and aerodynamics.
- Executed wind-box/bench tests on scale vehicle models; compared energy performance across body variants and control conditions.
- Built a Python pipeline that cut manual analysis time by 20%; produced plots for design reviews.
- Assisted in laboratory research planning and data analysis, strengthening experimental and technical documentation skills.

HONORS & AWARDS

- Distinction (Top 2.5%) in the 2023 Euclid Contest
- National Gold in the 2023 AAPT Physics Bowl Contest

PUBLICATION

- *Towards Intelligent Multi-Contact Planning for Humanoid Robots: A Capability-Oriented Survey (under review)*

TECHNICAL SKILLS

- **Programming:** Python, C/C++; MATLAB(foundational) , Verilog , ASM, JavaScript/TypeScript, SQL
- **Frameworks/Tools:** PyTorch, React, Supabase/PostgreSQL, Git
- **Embedded & E-Hardware:** Arduino/ESP32, PWM, sensor integration (LDR, ultrasonic, LiDAR/IR, bumper/cliff), motor drivers, MPPT, DC-DC converters
- **CS Fundamentals:** Data structures & algorithms (foundational)
- **Languages:** English (fluent), Chinese (native)